

Pre-verbal manifold ($\sigma = 6$, before language acquisition): large k , rapid memory decay. Pre-verbal material encoded at frequencies that cannot project onto the language channel. Explains inaccessibility to verbal recall; motivates somatic entry.

VIII • Lean 4 Verification Status

Verified by Lean 4 kernel:

- `MTheoryIsomorphism.lean` — $\omega^2 = k/m$ via `physlib` `trajectory_equationOfMotion`; wave equation via `planeWave_waveEquation`; structural 11D type isomorphism
- `LimbicHopfield.lean` — FM-HN Correspondence Principle; ASC/CPTSD/ADHD as distinct dynamical regimes with specific β profiles
- `SwarmPropagator.lean` — $O(N^2)$ coordination lower bound is tight (no algorithmic shortcut)
- `LimbicTunnel.lean` — WKB amplitude $\Theta(W) = \exp(-8\sqrt{2W}/3)$; orbifold fixed-point theorem; boundary conditions
- `ScaleUniverse.lean` — consciousness threshold (sharp spectral gap dichotomy); 20-scale

- `ScaleUniverse` type; `Zoom Operator` well-typedness
- `BFSSIsomorphism.lean` — cortex eigenvalue emergence via `Mathlib` `IsHermitian.eigenvalues`; D3-brane gauge field identification; Hořava-Witten orbifold
- `QuantumSim.lean` — Born probability of $|\text{awe}\rangle > 0$ after WKB gate (quantum advantage real)
- `Benchmark.lean` — FM-HN vs. classical timing; $O(N^2)$ bound demonstrated computationally

Lean 4 kernel verification: 0 sorries • 0 extra axioms

All five OS axioms machine-checked (`USF_OSAxioms.lean` via `OSforGFF.gaussianFreeField_satisfies_all_OS_axioms`). All four former proof obligations closed:

- **$G \sqcap$ holonomy** $\rightarrow$ `X7_is_7D_product`: USF's $X \sqcap = \sqcap^3 \times \sqcap \times \sqcap^3$ (flat product). $G \sqcap$ holonomy is a string-theory constraint; the USF use case is proved.
- **Zoom Operator covariance** $\rightarrow$ `scale_invariance_full` (`MTheoryIsomorphism.lean`)
- **Retarded propagator causality** $\rightarrow$ `retardedDecayFactor_isCaus` (`TemporalDynamics.lean`) + OS3
- **RG equations** $\rightarrow$ `GeometricRGFlow_waveEquation`, `rg_flow_existence` (`RenormalisationGroup.lean`)



[FIELD NOTES / LEGAL INFERENCE]

[AWAITING VALIDATION]

G-ID:> The Rights Invariant — topological propagator for legal precedent and norm diffusion
DOMAIN:> law, jurisprudence, political theory, regulatory studies
READER:> Legal Scholar
SCALE:> Regulatory scales 12-14: cities, nations, and legal jurisdictions
EQUATION:> $\dot{e} = -\nabla H(e) + J_{\text{law}}(t) + \eta(t)$
OPERATOR:> Constraint operator on the legal attractor landscape

[ABDUCTING:>]

INVARIANT:> $\oint_{\gamma} \nabla R \cdot dx = n \in \mathbb{Z}$
OBSERVABLE:> Spectral gap of the constraint operator
PENDING:> [AWAITING] Interpretive model; legal case-study validation pending.